Finite Groups and Their Representations — Exercise Sheet 10

The Gelfand–Tsetlin algebra

Variant: Questions only

Exercise 1 (Direct commutation). In $\mathbb{C}[S_n]$ use $X_1 = 0$ and $X_k = \sum_{a<k}(a\,k)$. Prove directly that the X_k commute, without invoking GZ(n). In the commutator of X_i and X_j for $i < j$, group the non-disjoint terms in triples of points. Prove that $\sum_{k=1}^{n} X_k = \tau_n$ is central. Include $n = 1$.

Exercise 2 (Conjugacy with one fixed label). Find a permutation fixing 6 that conjugates $\sigma = (1\,6\,4)(2\,3) \in S_6$ to its inverse. Use Lecture 10's classification by cycle type and the cycle through the fixed label to count the S_{n-1}-conjugacy classes in S_n for every $n \geq 2$. Evaluate the count for $n = 6$. Write $p(t)$ for the number of partitions of t, with $p(0) = 1$. Deduce the dimension of $\mathbb{C}[S_6]^{S_5}$.

Exercise 3 (GZ lines in the standard representation). For each $j \geq 2$, let $W_j = \{x \in \mathbb{C}^j : \sum_{a=1}^{j} x_a = 0\}$ have the natural S_j-action; you may take its irreducibility as given. Here W_j denotes this zero-sum module, independently of the isotypic chain notation in Lecture 10. Let $n \geq 3$. For $2 \leq k \leq n$ define

$$v_k = e_1 + \cdots + e_{k-1} - (k-1)e_k.$$

Prove that the lines $\mathbb{C}v_k$ are the GZ lines of W_n along the standard subgroup chain. Find every joint weight of $X_1, \ldots, X_n$ on W_n, including the full list for $n = 5$. Compute the scalar by which τ_n acts on W_n and compare with the central-character formula of Lecture 9. Do not use tableau labels.

Exercise 4 (A path projection and an isotypic projection). In S_4 let $W = W_4$ be the zero-sum module, and let E be the S_3 plane. The extra group subscript specifies which group algebra contains an idempotent. For the path $\gamma = (\mathbf{1} \to \mathrm{sgn} \to E \to W)$, write $P_\gamma = e_{\mathrm{sgn},S_2} e_{E,S_3} e_{W,S_4}$ using the central-idempotent formula of Lecture 7 as an explicit product of three factors. Identify its image on W and its rank on the left regular module $\mathbb{C}[S_4]$. Compare with e_{W,S_4}. Explain why summing P_γ over the paths ending at W gives e_{W,S_4}.

Exercise 5 (Adjacent transpositions on the explicit lines). Use the vectors v_k of Exercise 3. For $2 \leq i < n$, compute the matrix of $s_i = (i\,i+1)$ on the ordered pair (v_i, v_{i+1}). Show that it fixes every other v_k, and give the action of s_1. Normalise the basis and compute the resulting two-by-two matrix. Verify $s_i X_i + \mathbb{1} = X_{i+1} s_i$ on this pair using the weights.

Exercise 6 (A quadratic central element). Prove directly in $\mathbb{C}[S_n]$ that

$$\sum_{k=1}^{n} X_k^2 = \binom{n}{2} e + \sum_{\substack{\sigma \in S_n \\ \sigma \text{ is a three-cycle}}} \sigma.$$

Deduce centrality, including $n = 1, 2$ with an empty three-cycle sum. Find its scalar on W_5 in two ways: from the weights in Exercise 3 and from the central-character formula and the fixed-point character.

Exercise 7 (⋆ Marked permutations and Olshanskii's theorem). Let $l, k \geq 1$, let $n = l + k$, and embed S_l in S_n by fixing all points of $M = \{l+1, \dots, n\}$. These points are called marked. Let S_M denote their permutation group and put $Y_a = \sum_{i=1}^{l} (i\, a)$ for $a \in M$. Prove the full centraliser statement

$$\mathbb{C}[S_n]^{S_l} = \langle Z(\mathbb{C}[S_l]), \mathbb{C}[S_M], Y_a \ (a \in M) \rangle = \langle Z(\mathbb{C}[S_l]), \mathbb{C}[S_M], X_{l+1}, \dots, X_n \rangle.$$

Classify S_l-orbits by the cyclic order of the marked points, the number of unmarked points in each intervening run, and the cycle type of the cycles containing no marks. Use induction on the number of moved unmarked points. This independent extension is estimated at 40 minutes.